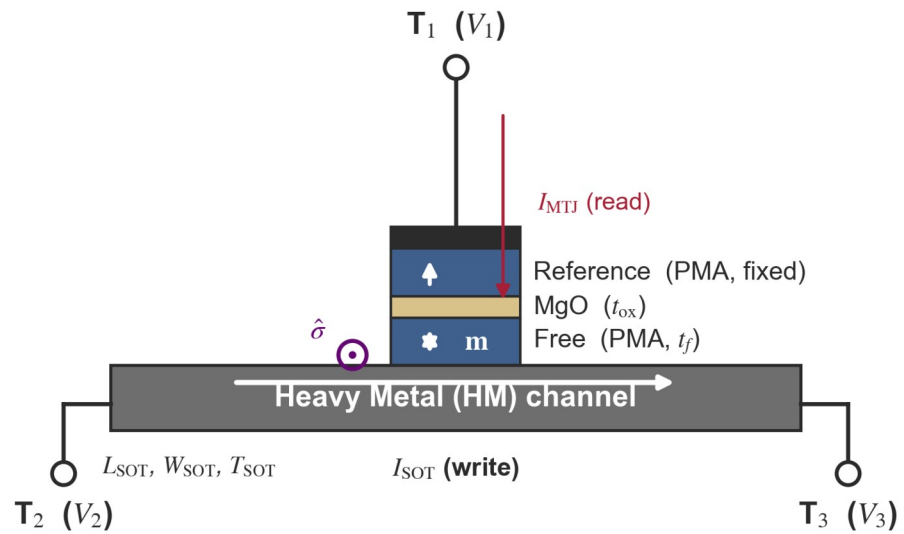


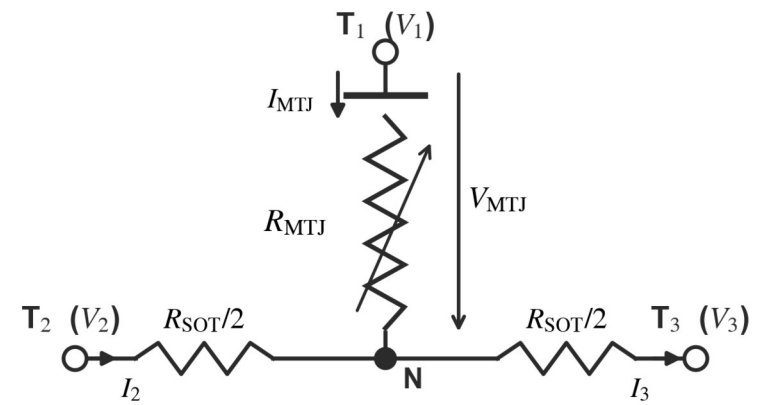
(a)

Physical device structure

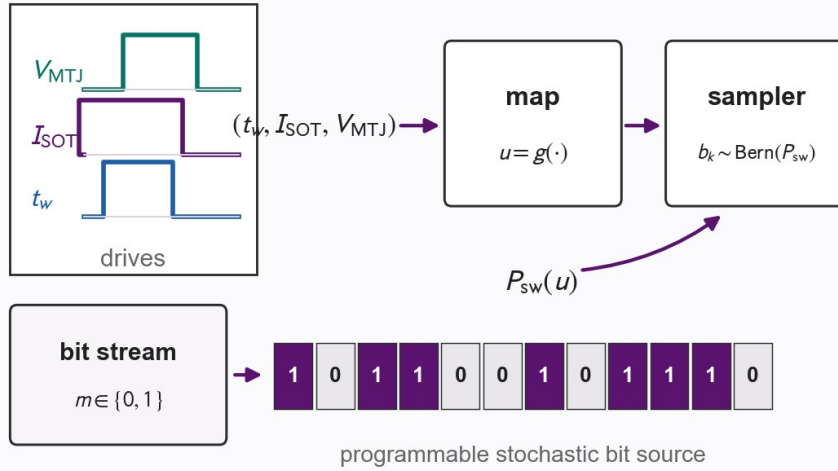


(b)

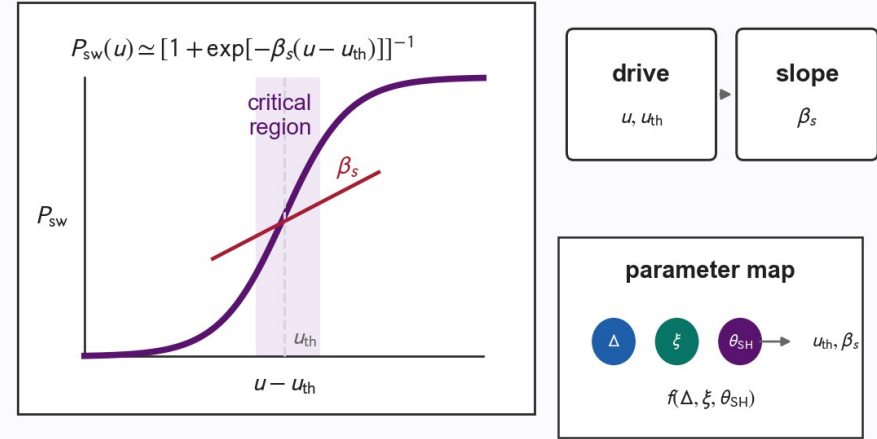
T-equivalent resistor network



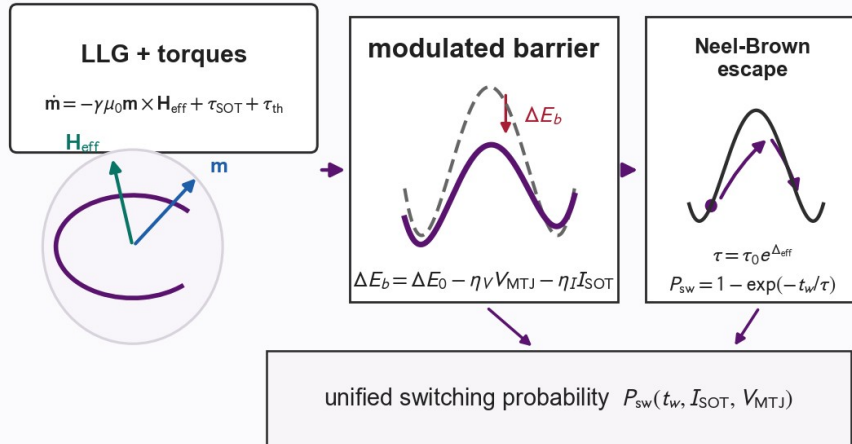
(a) Computational layer



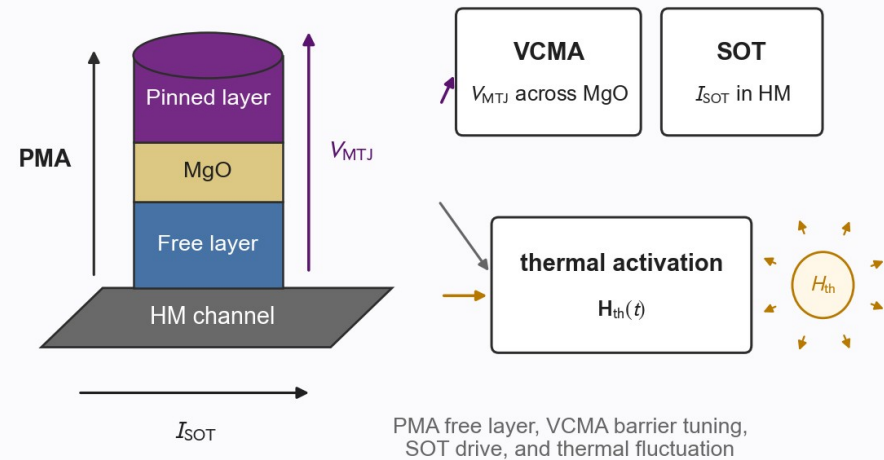
(b) Behavioral layer



(c) Physical-model layer

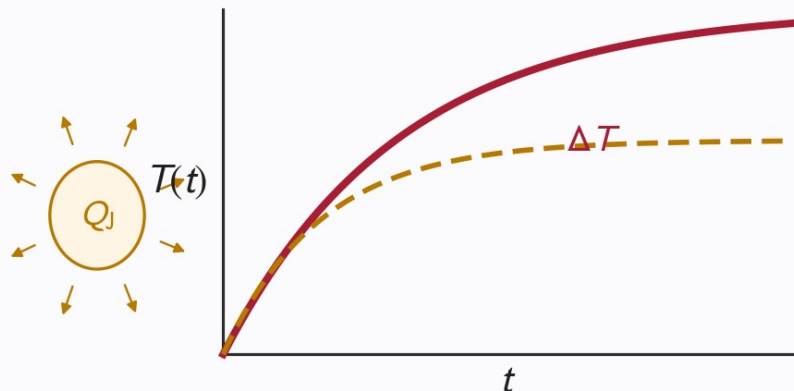


(d) Device layer



(a) Self-heating

$$C_v \dot{T} = \lambda \nabla^2 T + Q_J$$



(b) T-dependent channels

materials

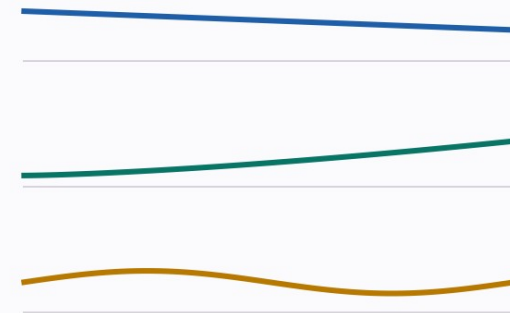
M_s, K_i, η

transport

R_{MTJ}

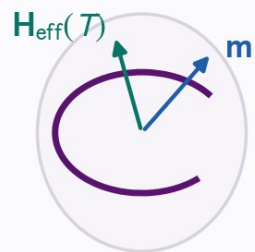
demag

H_D



T -corrected inputs

(c) Stochastic LLG



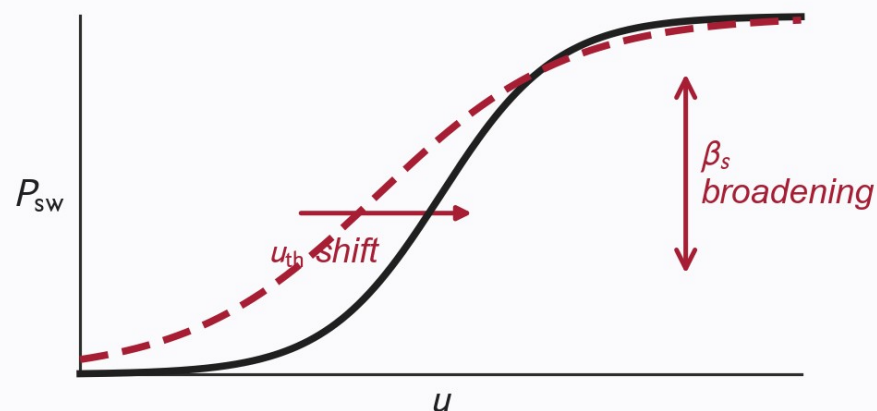
stochastic LLG

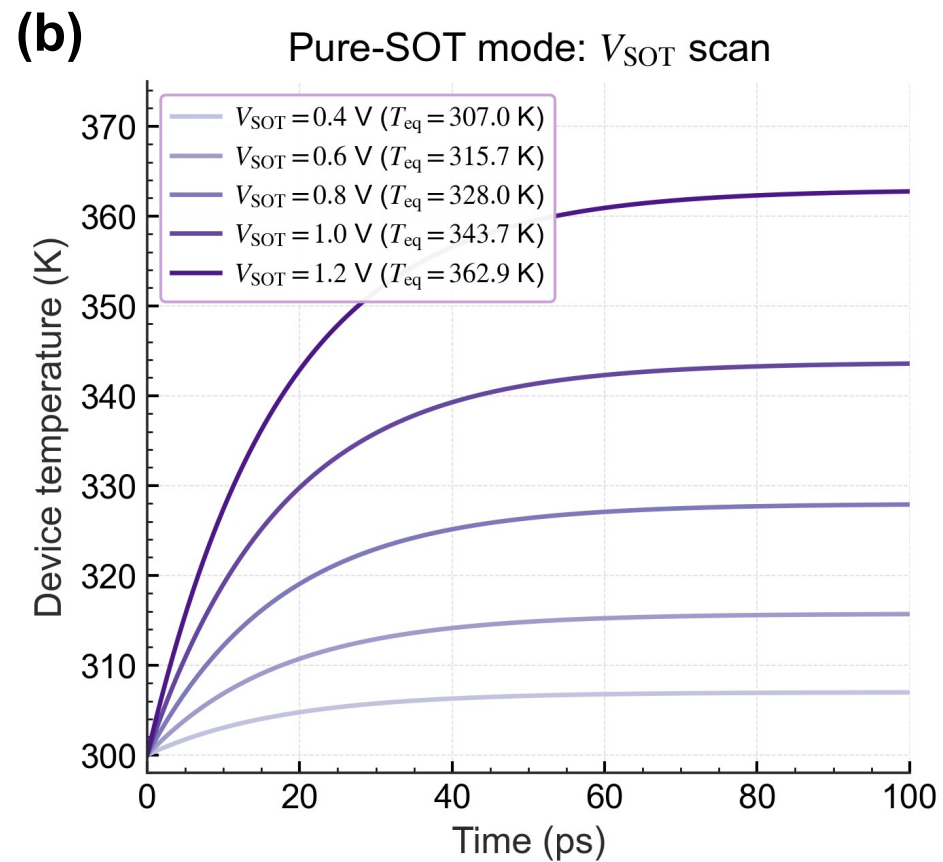
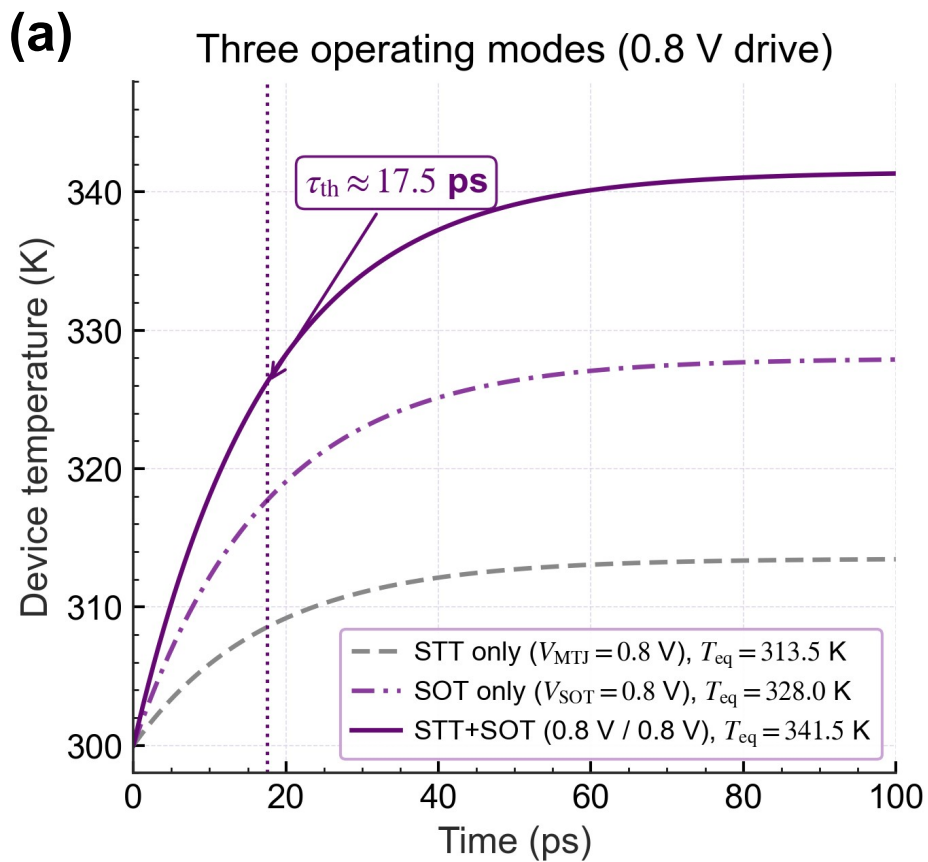
$$\dot{\mathbf{m}} = -\gamma \mu_0 \mathbf{m} \times \mathbf{H}_{\text{eff}} + \tau_{\text{SOT}} + \tau_{\text{th}}$$

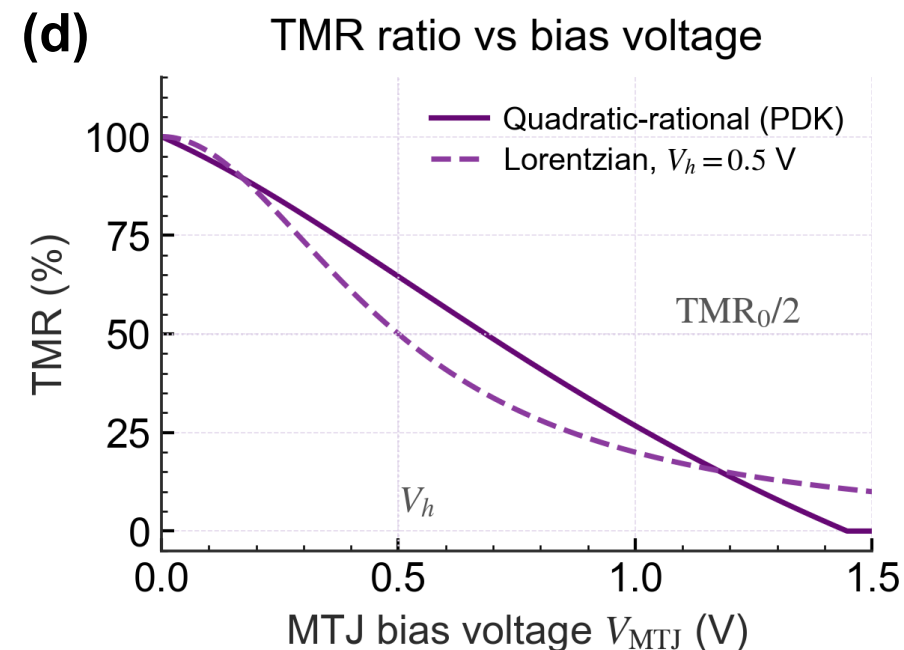
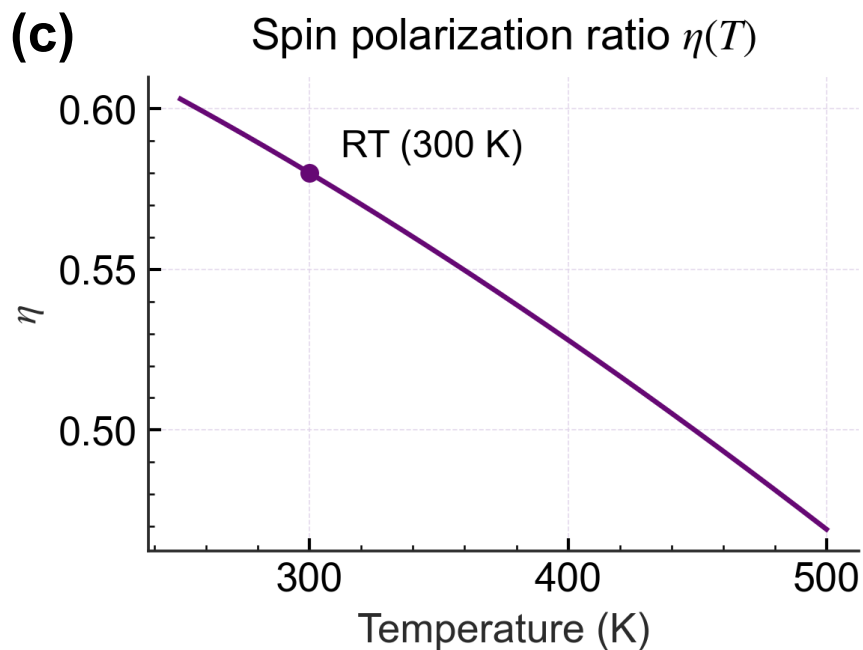
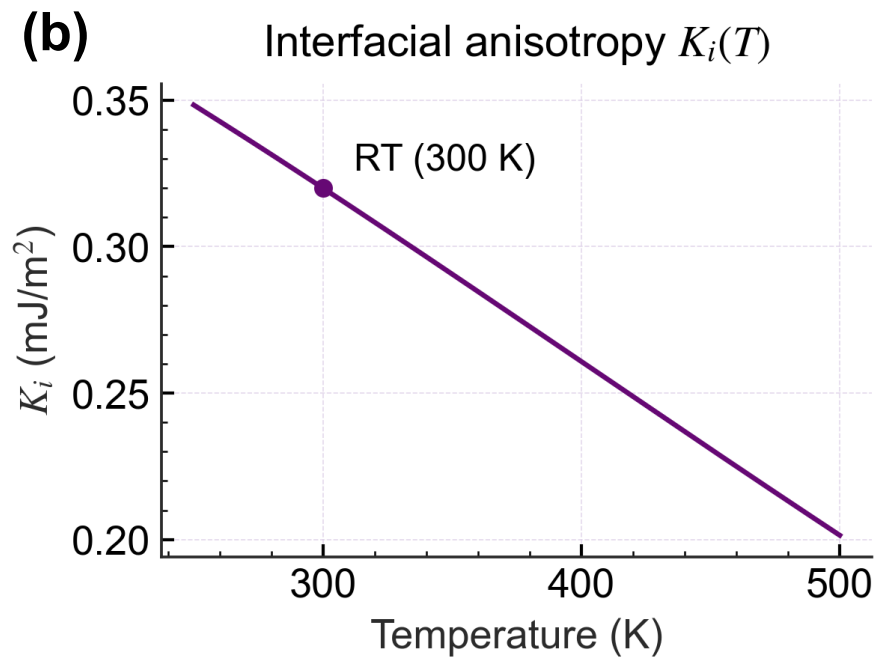
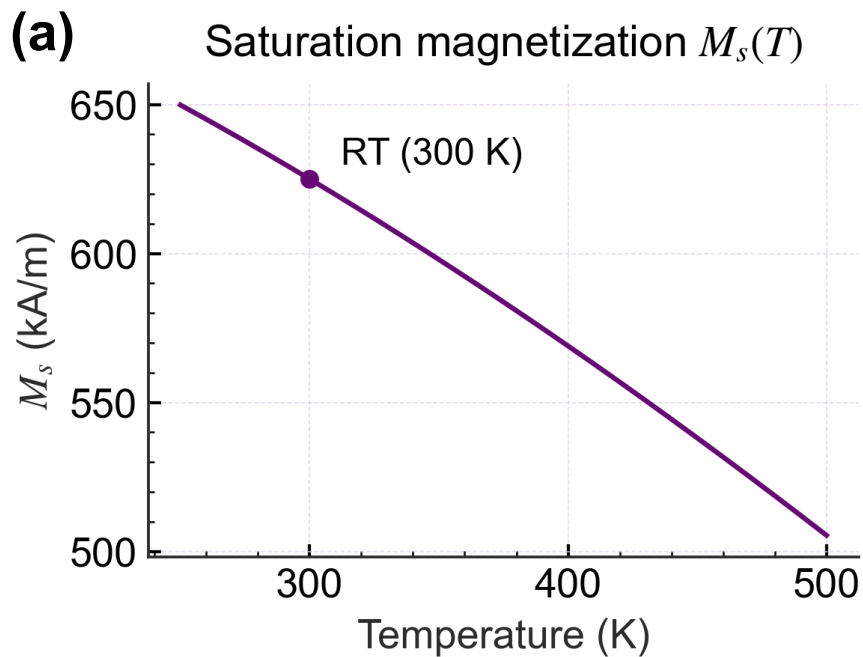
Δt
update

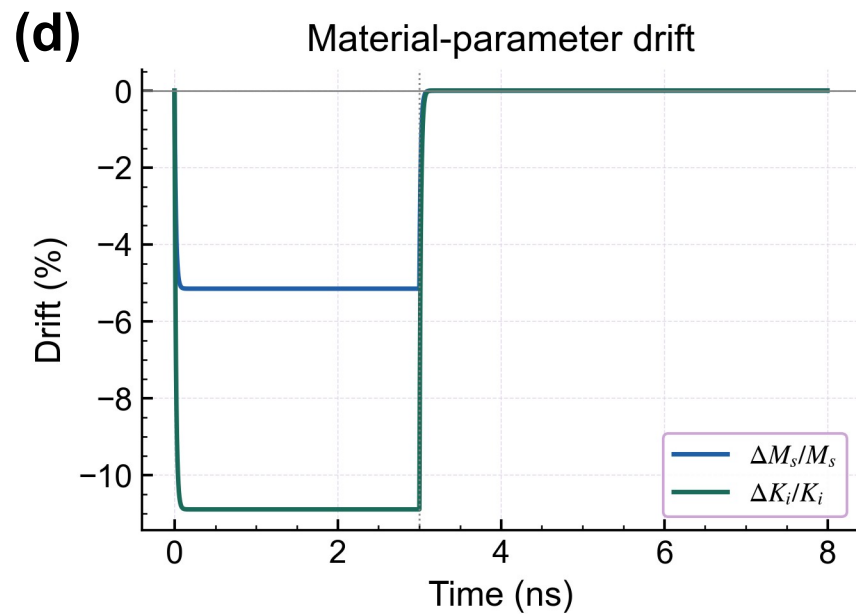
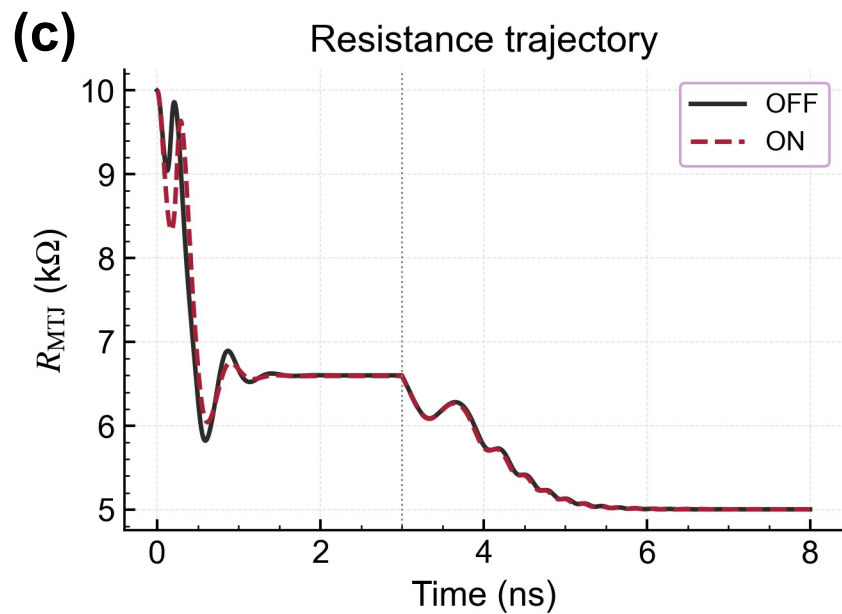
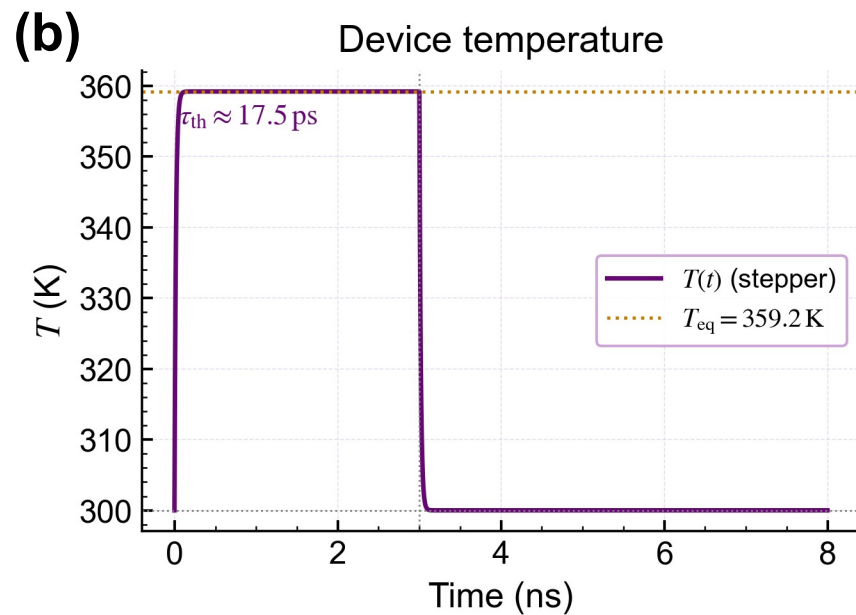
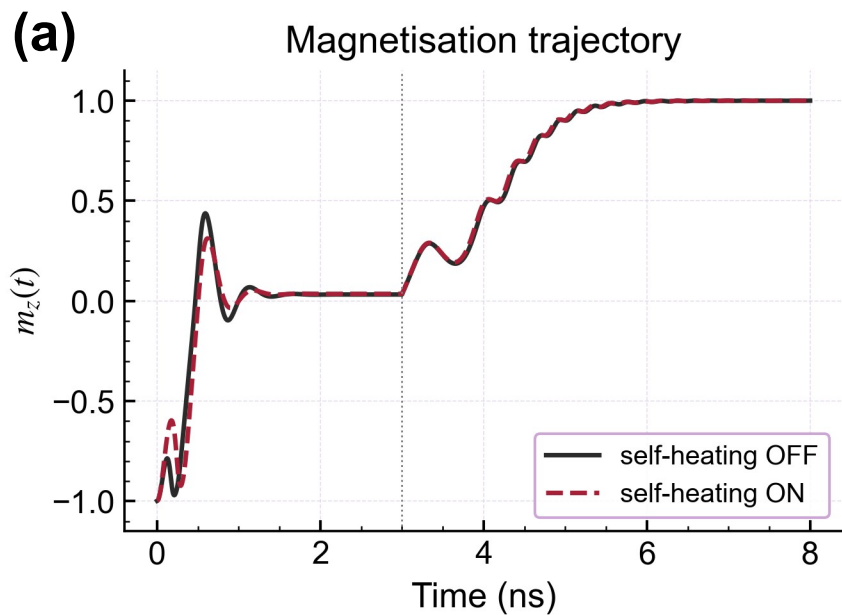
t_w
sample

(d) P_{sw} response









User interfaces and I/O

CLI / Python API

install
pip

CLI
sweeps

Python API
scripts

Experiment scenarios and configuration

cases / configs

standard cases

SOT baseline
optimized VGSOT

VCMA-assisted
SER Monte Carlo

configuration

J, V, t_w, T

Physical kernels and core modules

single-step update loop

dynamic_switching

$$\mathbf{m}_{n+1} = \mathcal{F}(\mathbf{m}_n, \mathbf{H}_{\text{eff}}, \mathbf{H}_{\text{th}})$$

anisotropy
field

stochastic
noise

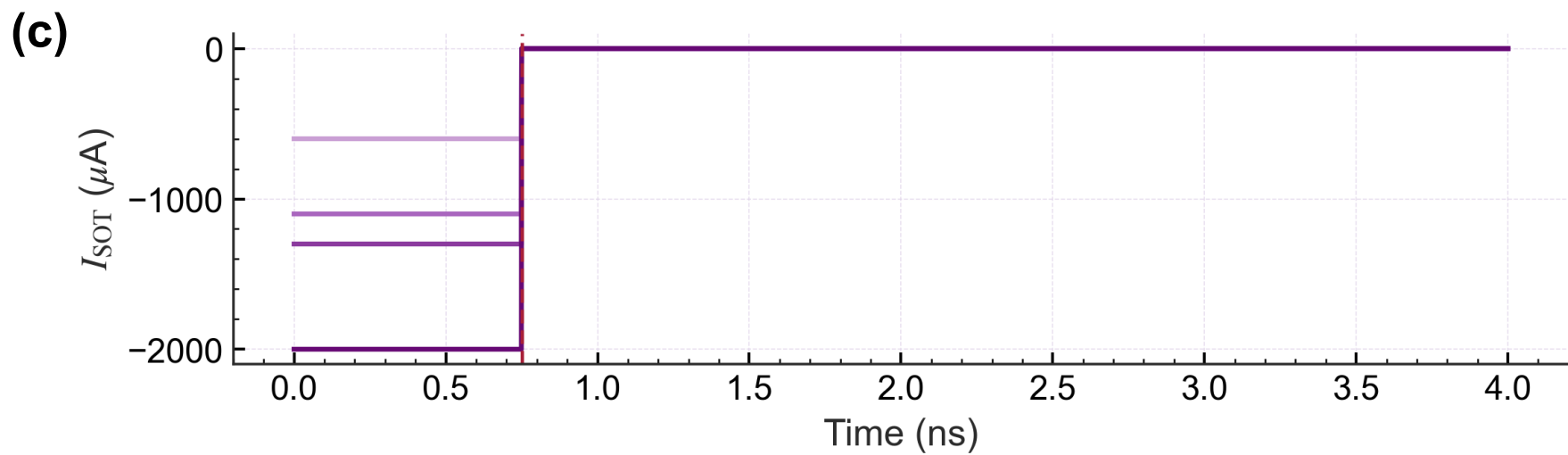
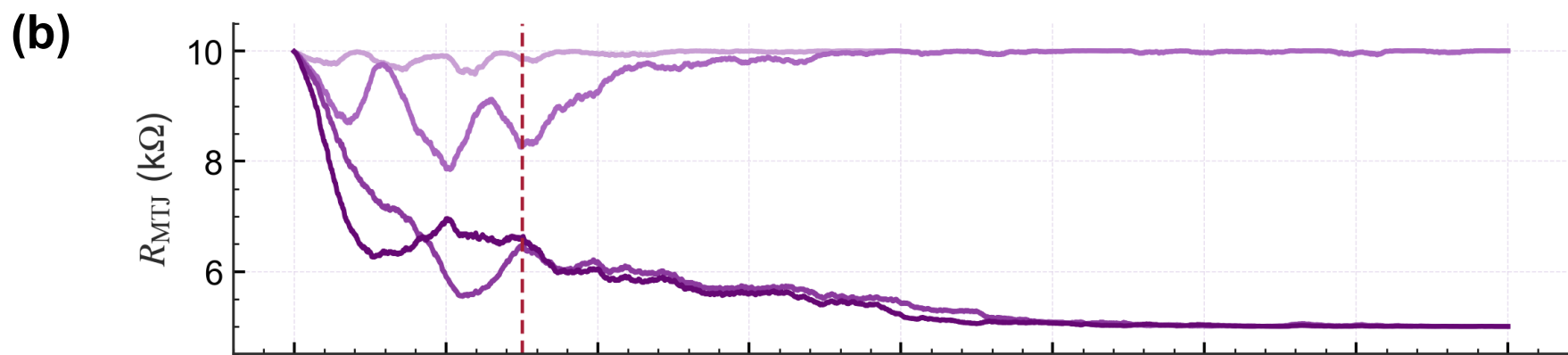
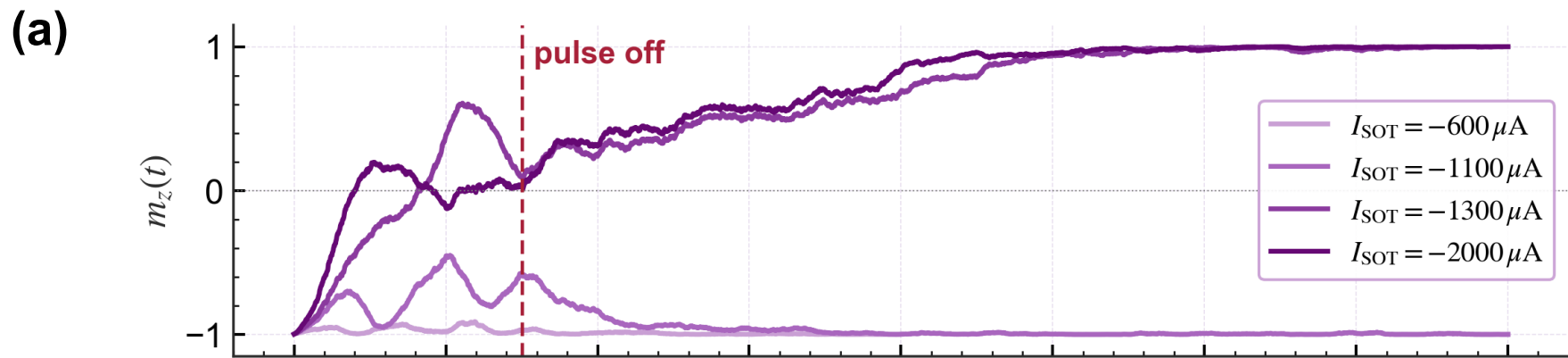
electronic
transport

tmr
resistance

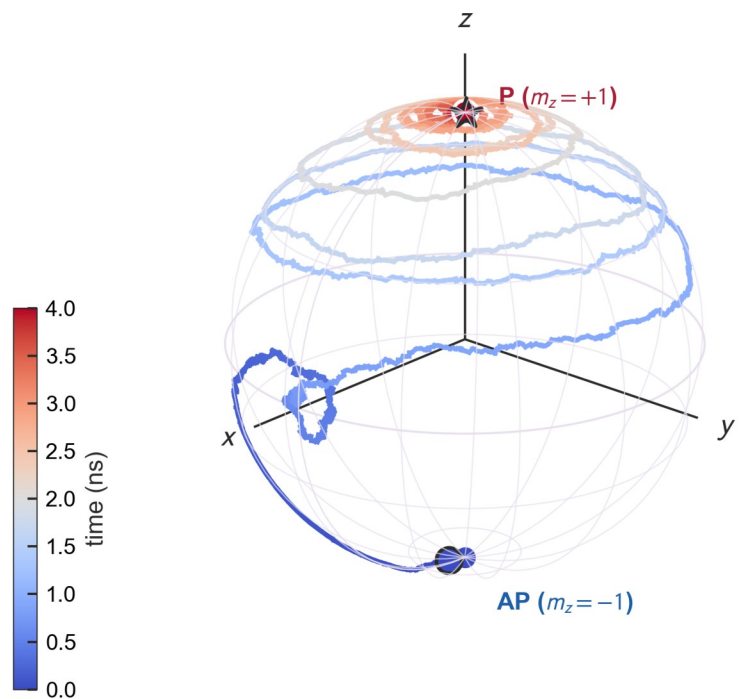
inputs
 J, V, t_w, T

single-step data path

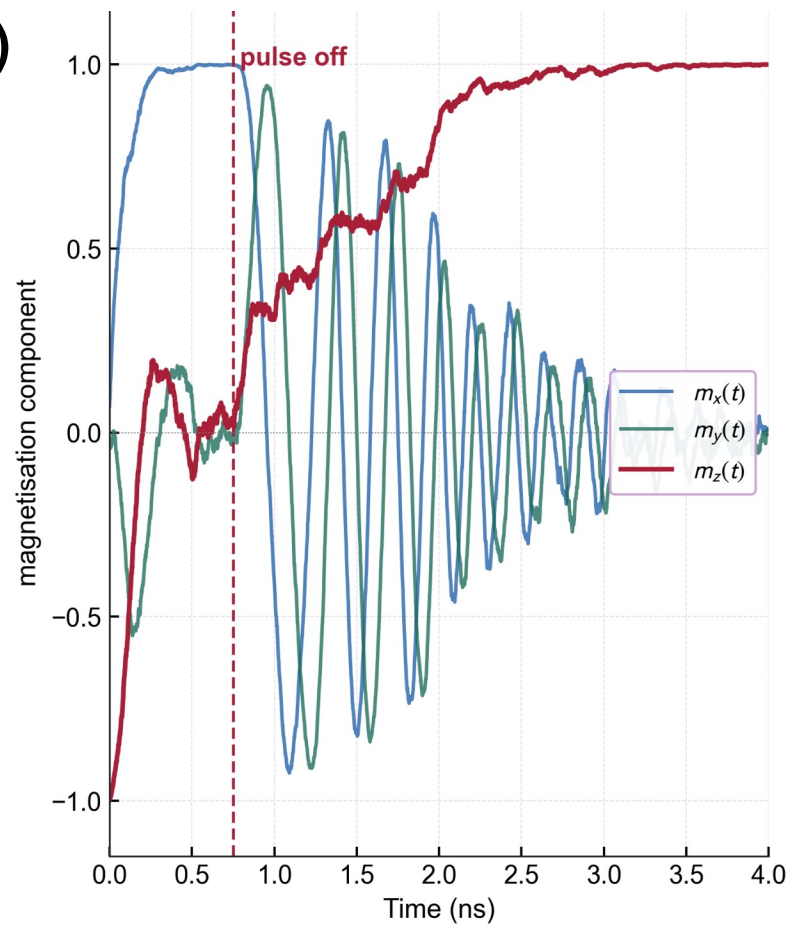
outputs
 $P_{\text{sw}} / \text{SER}$



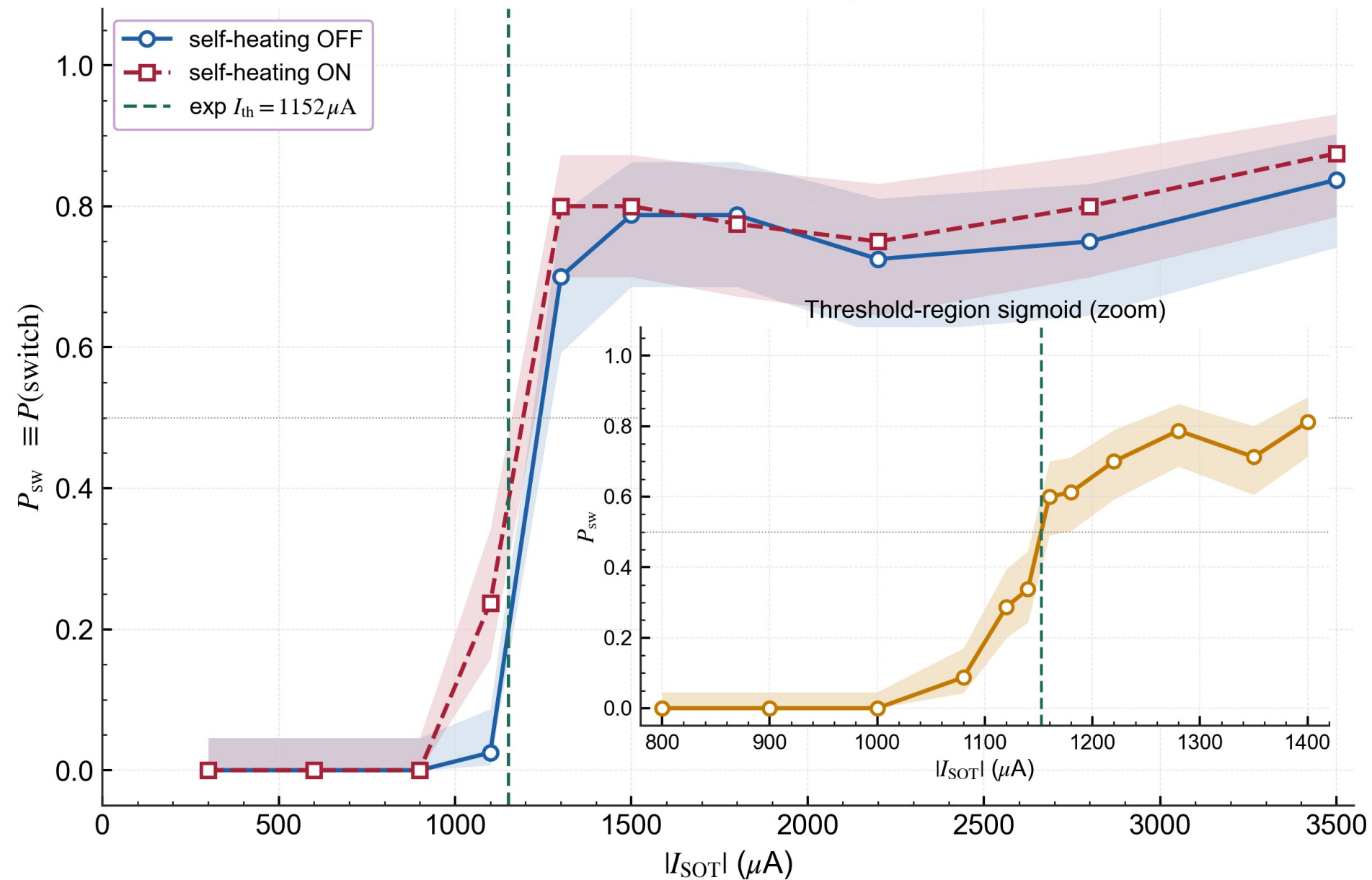
(a)



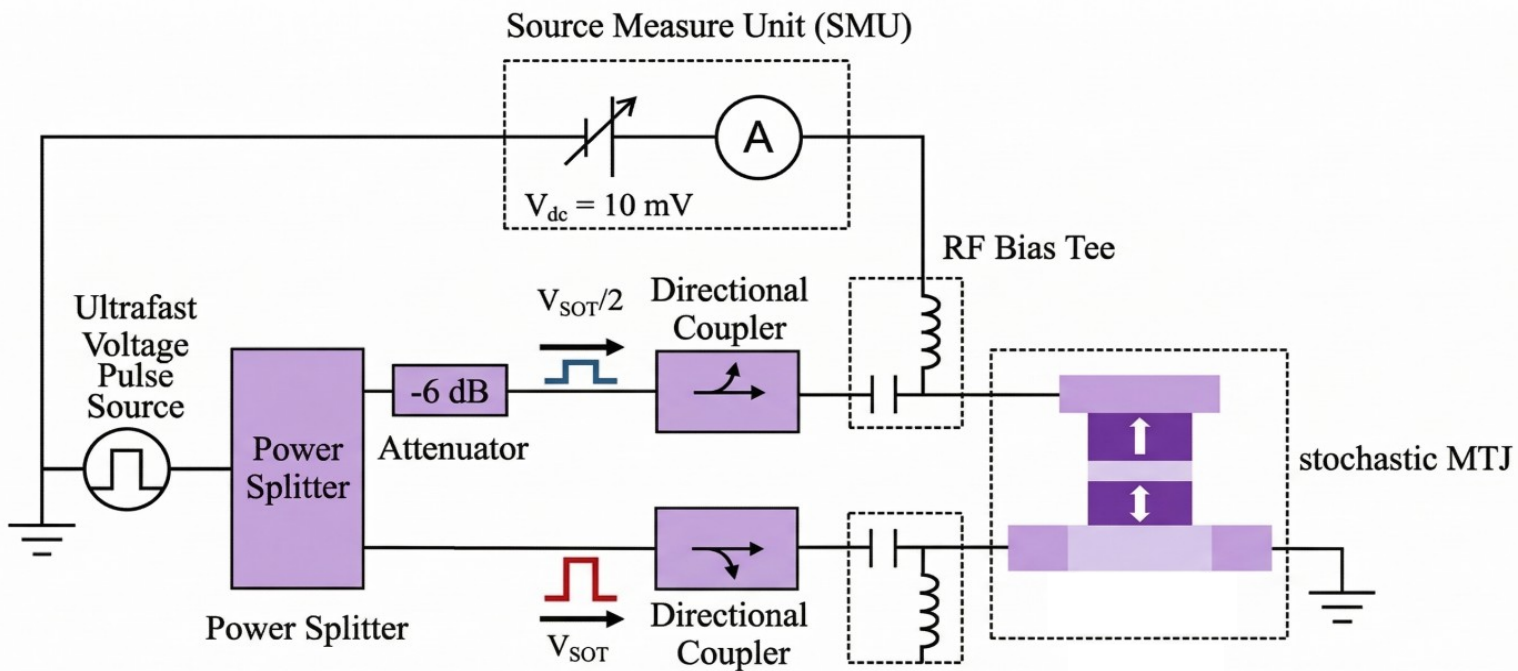
(b)



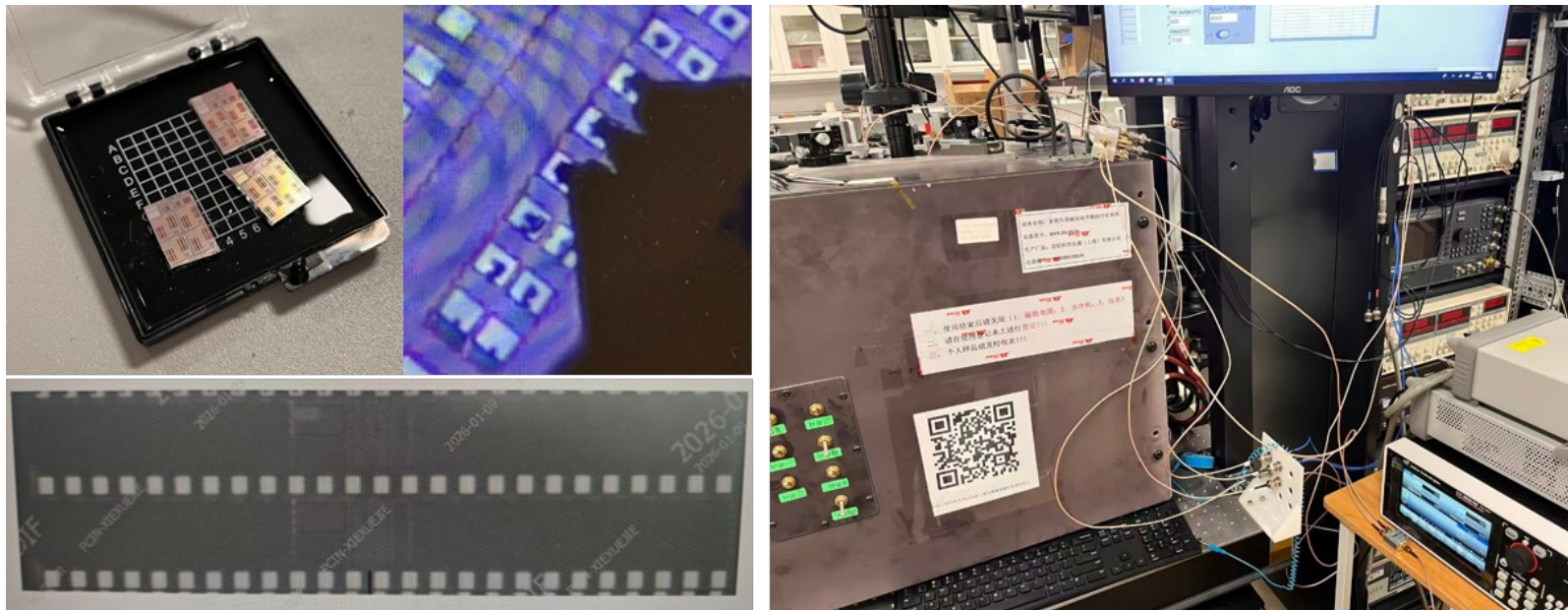
vgsot-sim Monte-Carlo P_{sw} sweep, $t_p = 0.75 \text{ ns}$, $V_{\text{MTJ}} = 0$

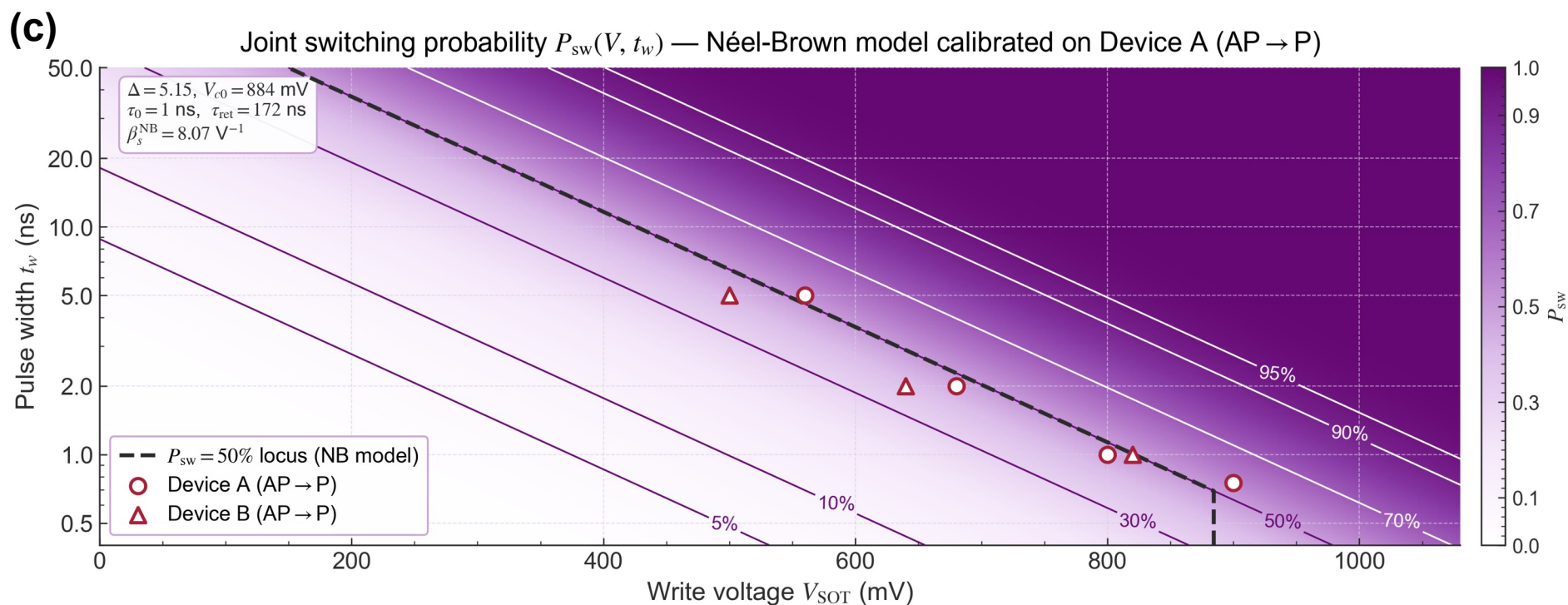
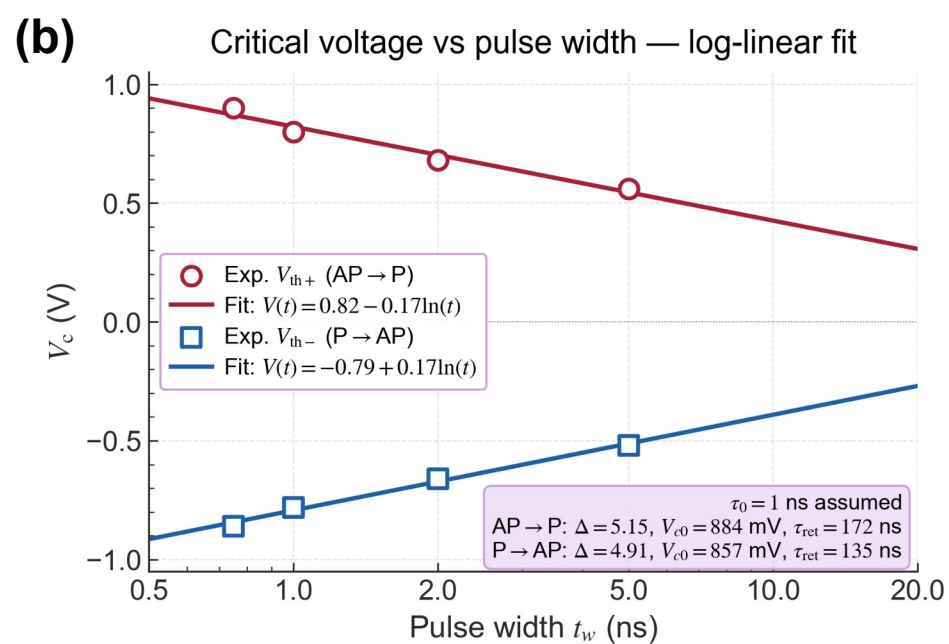
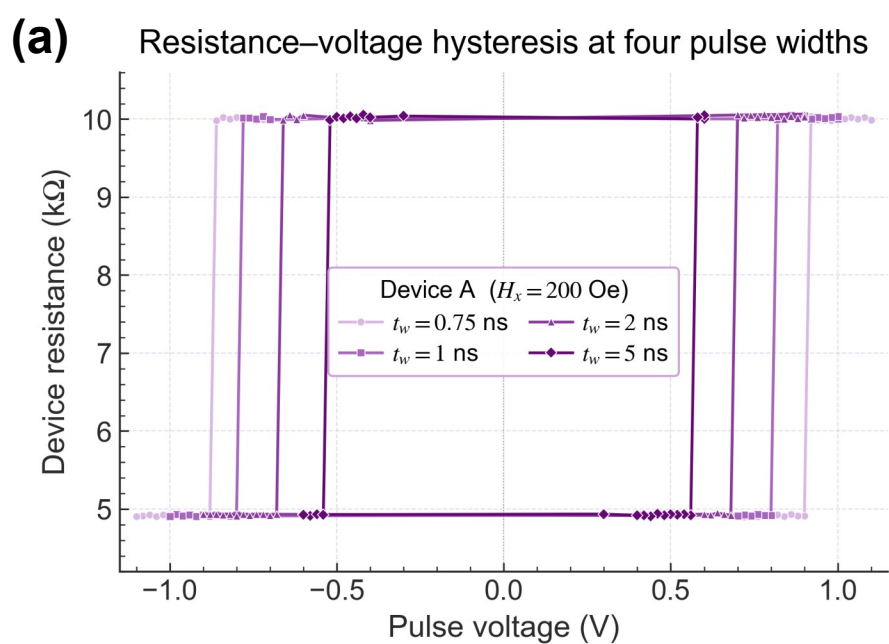


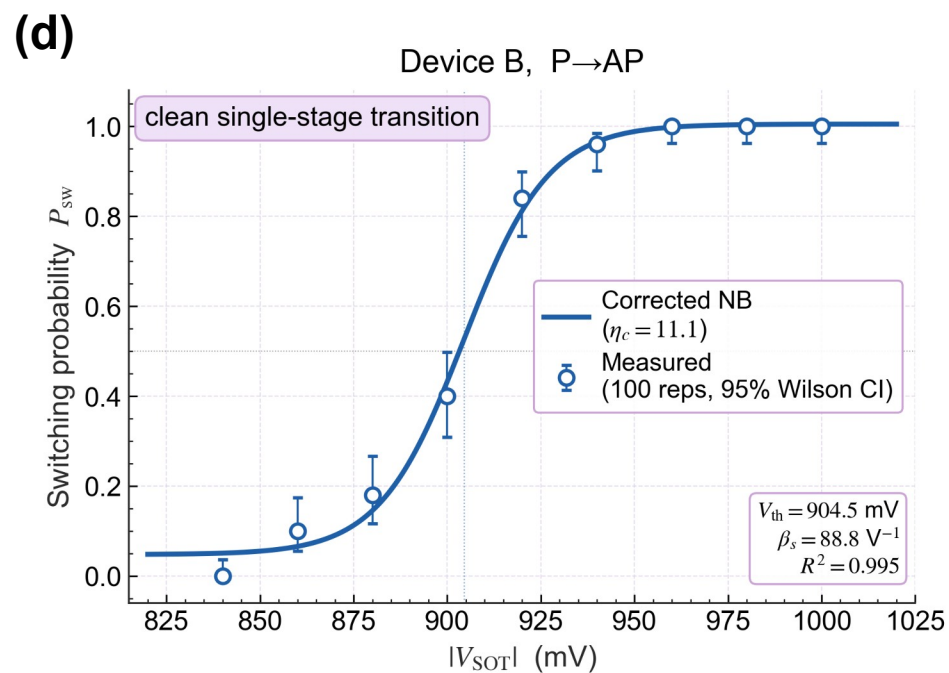
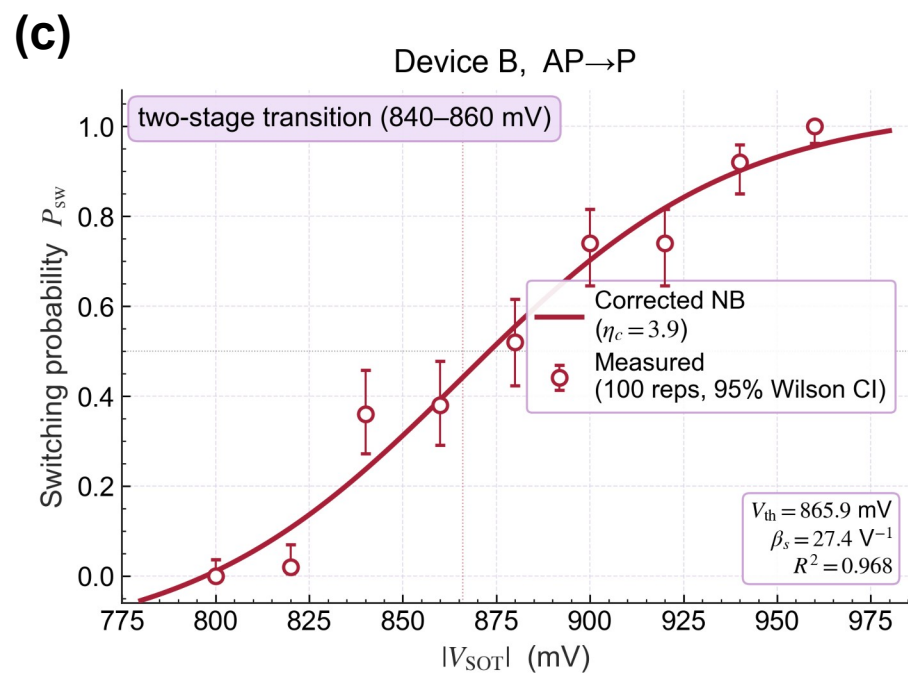
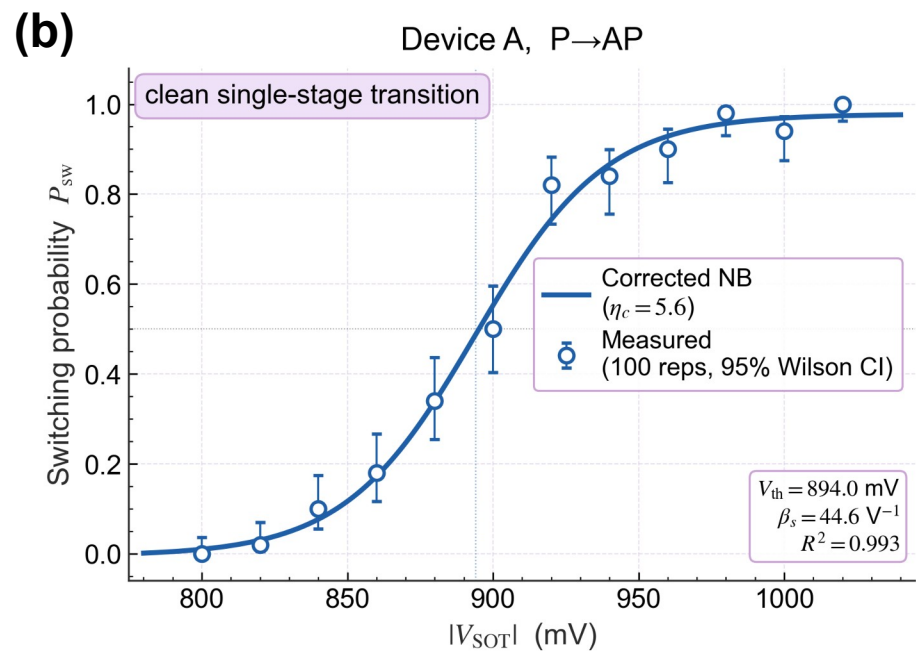
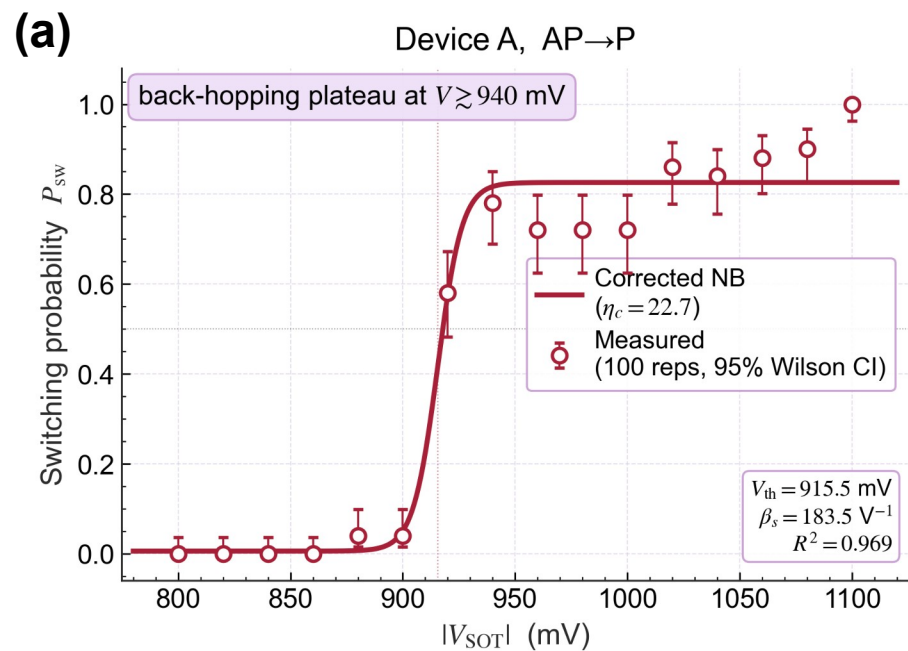
(a)

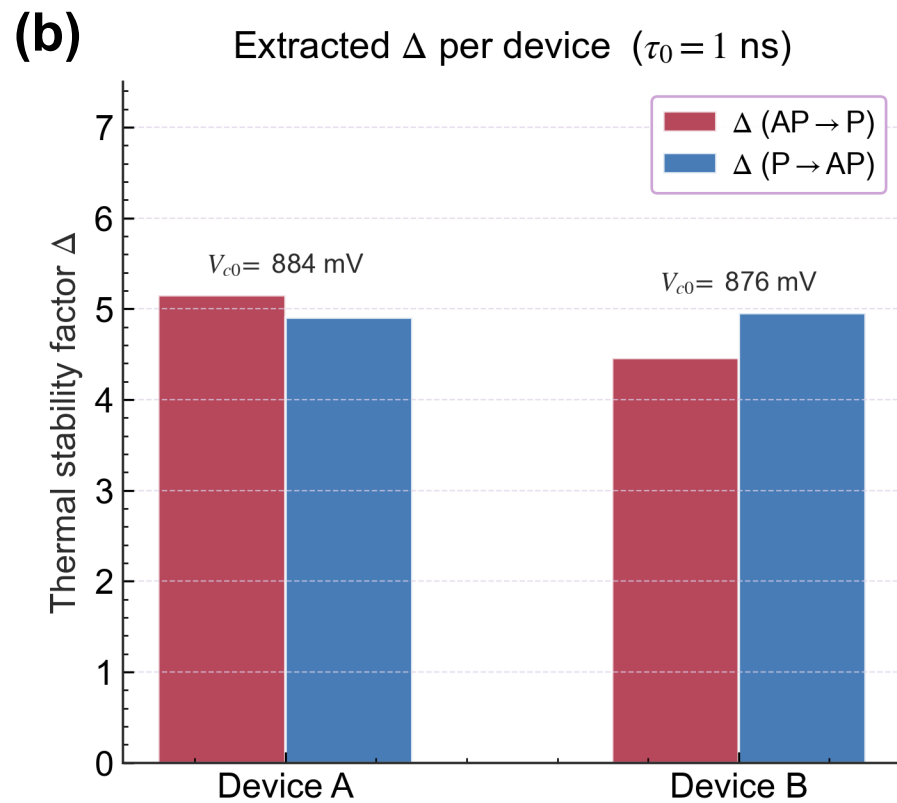
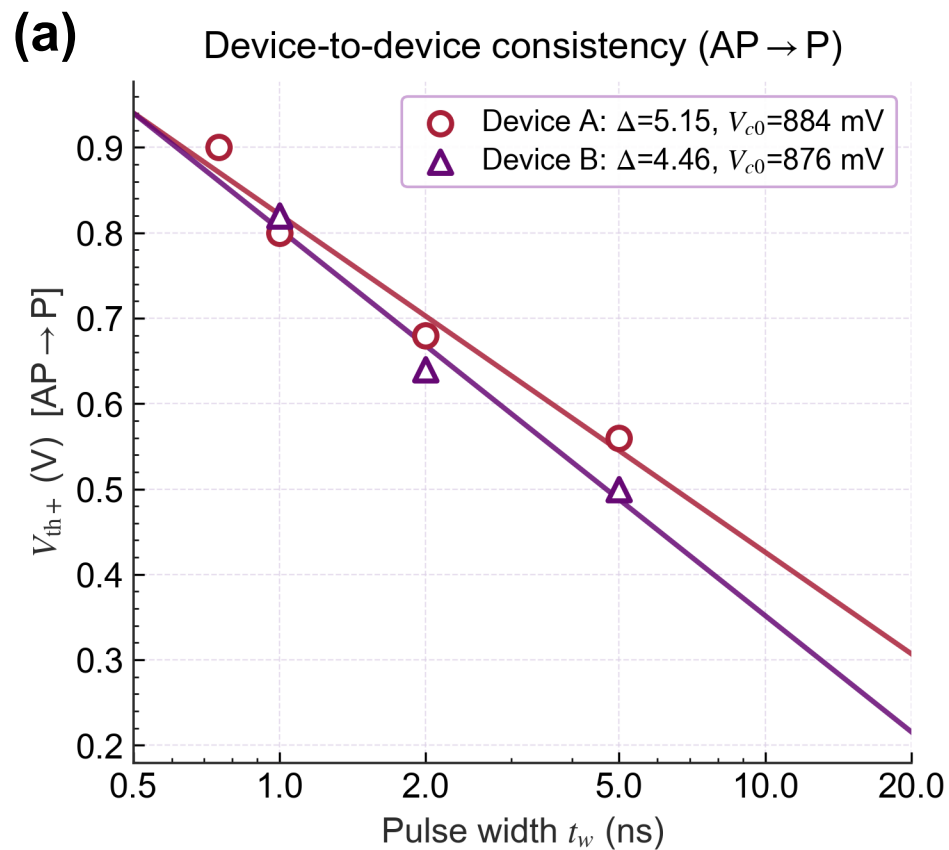


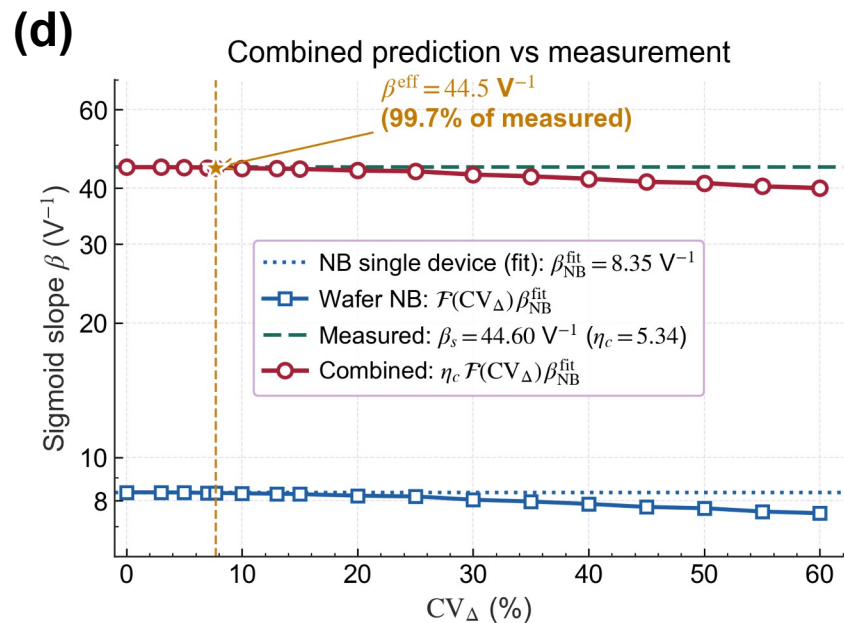
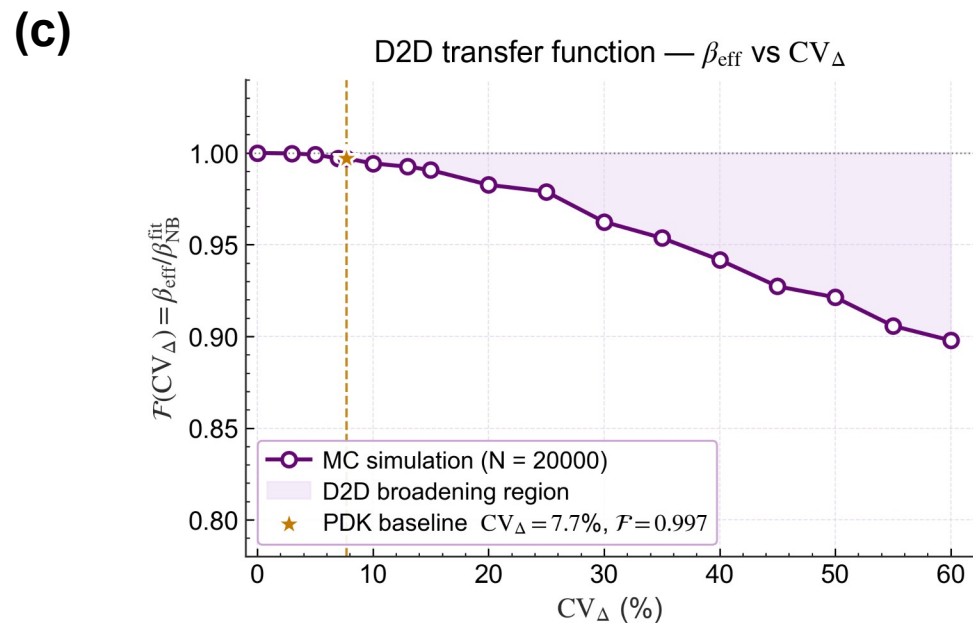
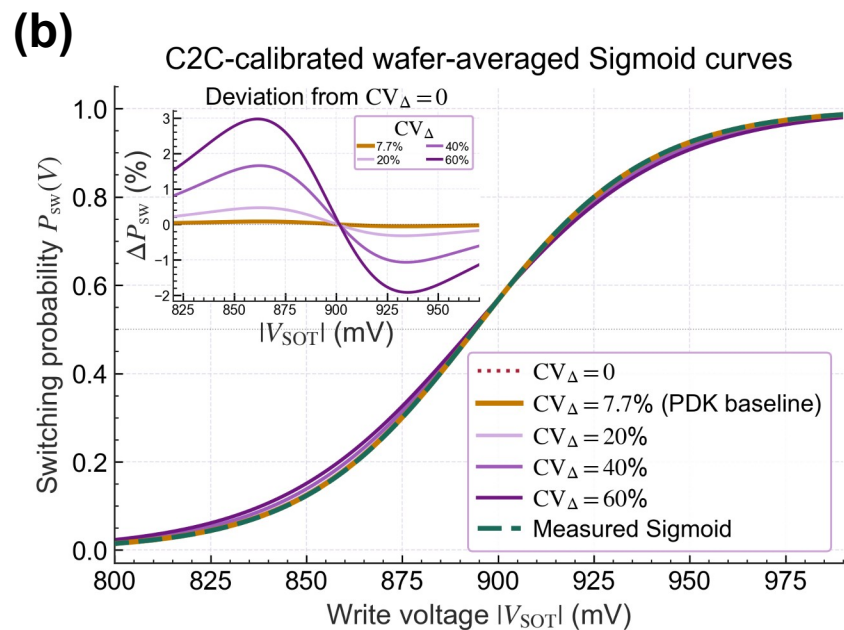
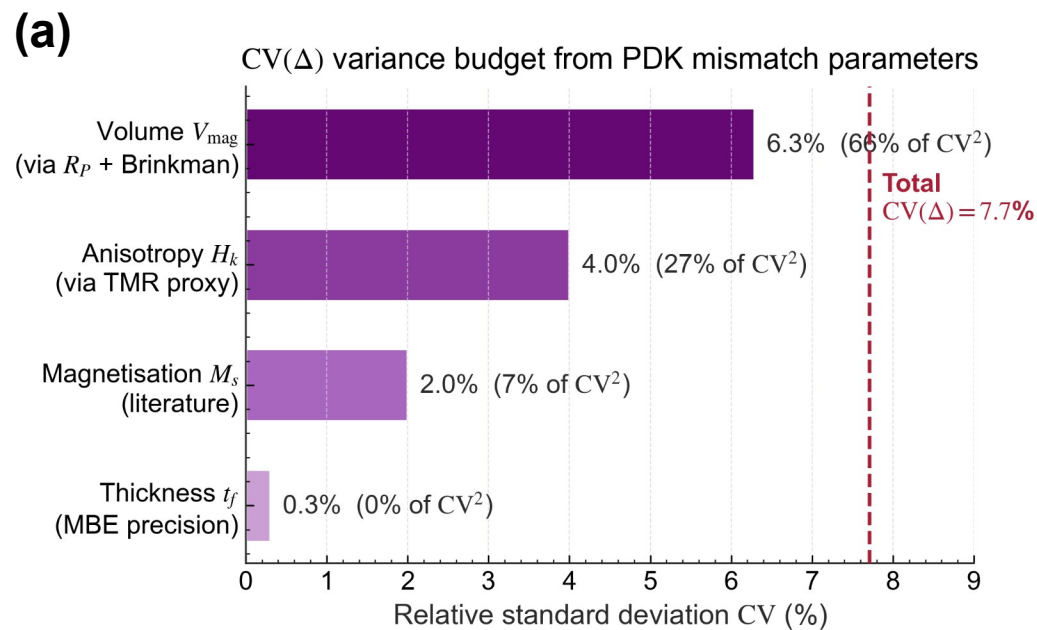
(b)



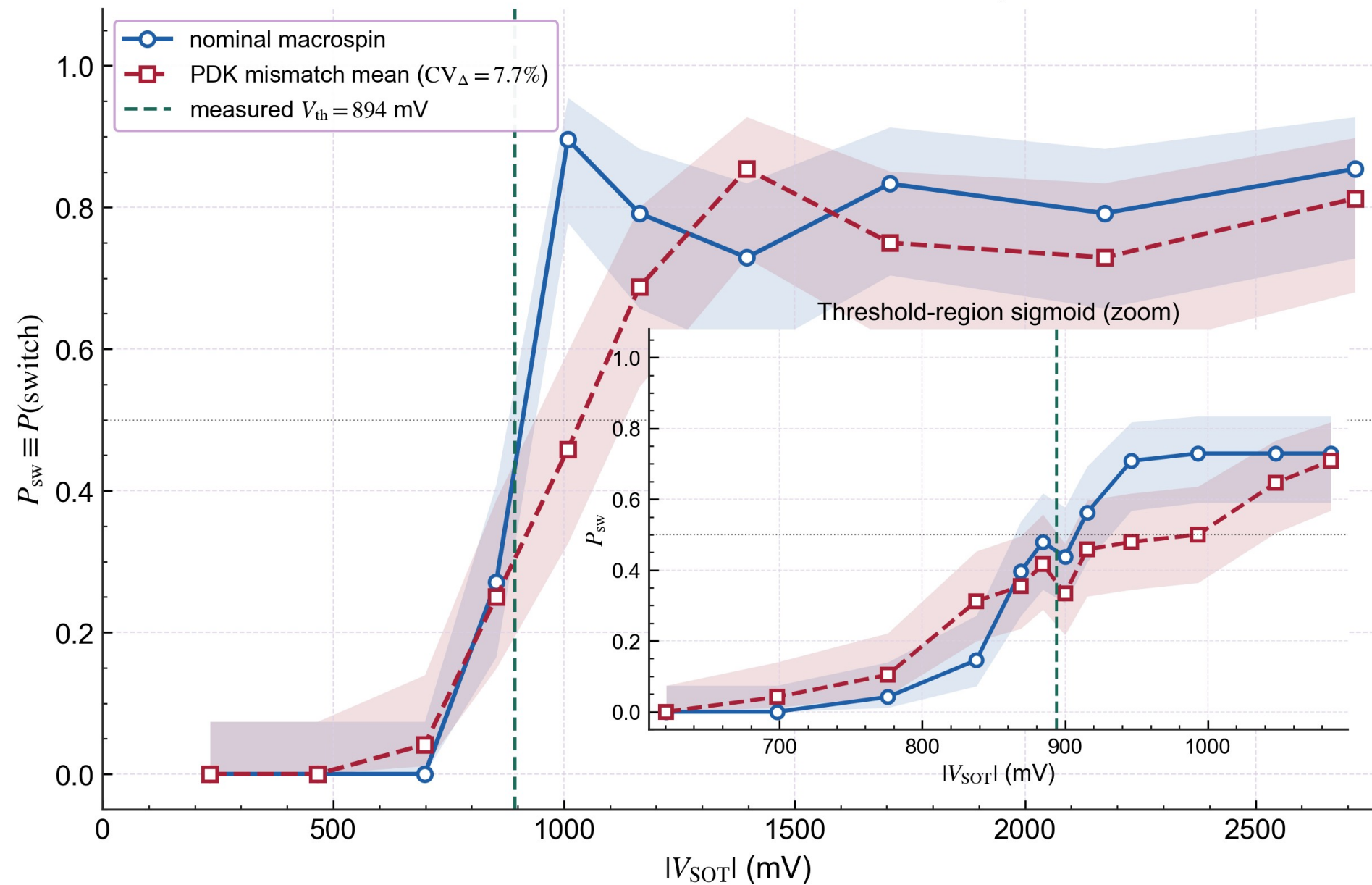


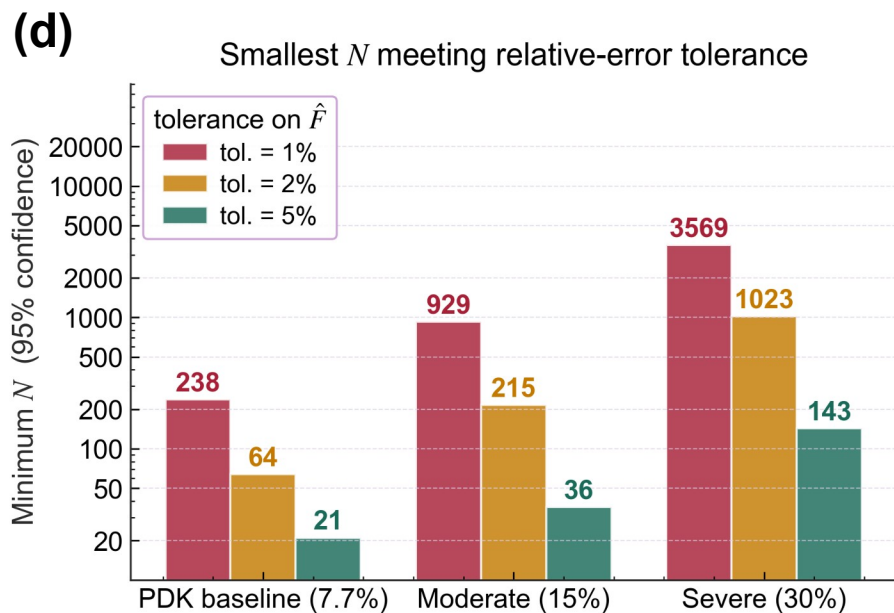
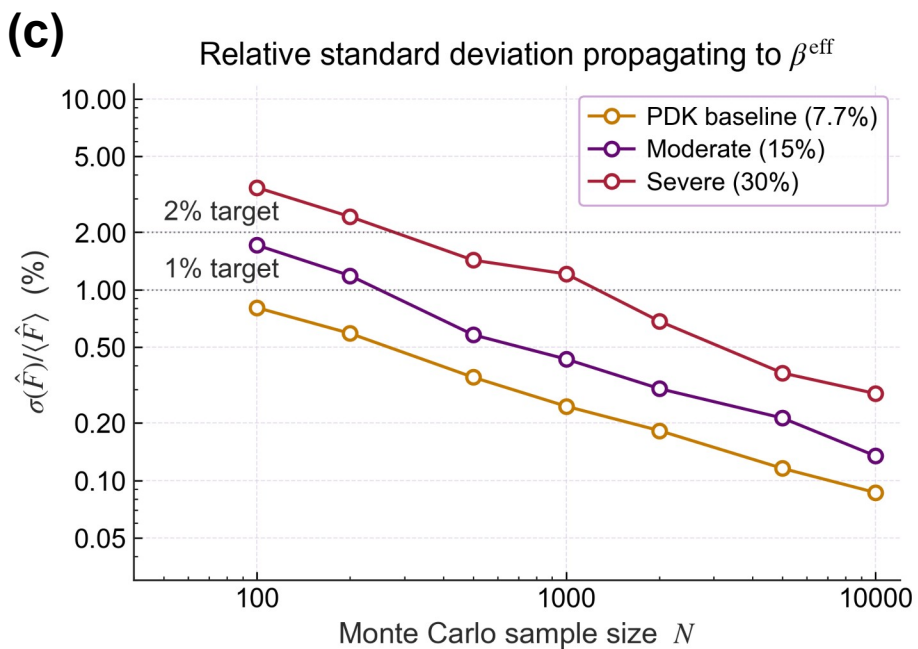
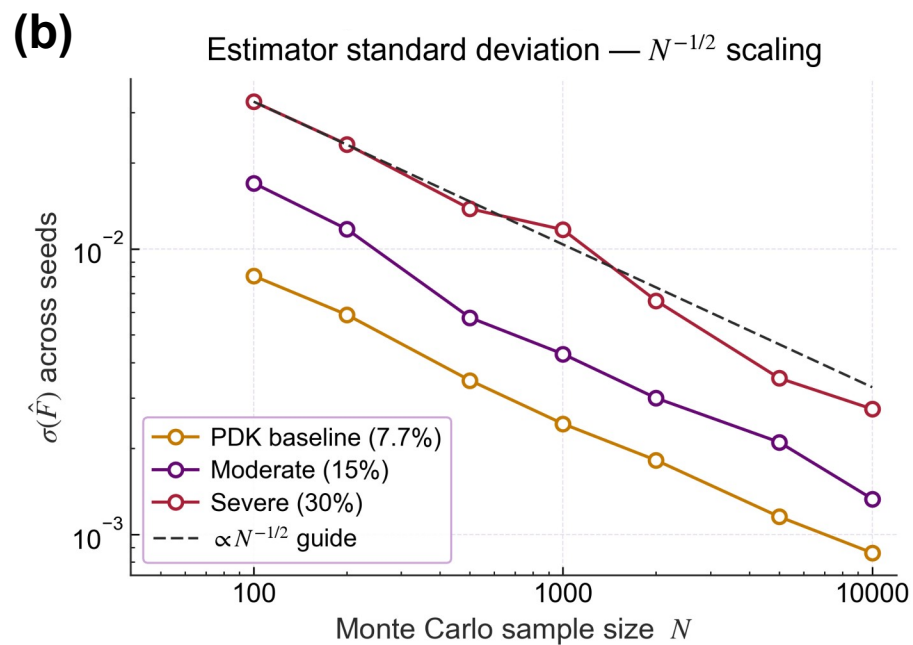
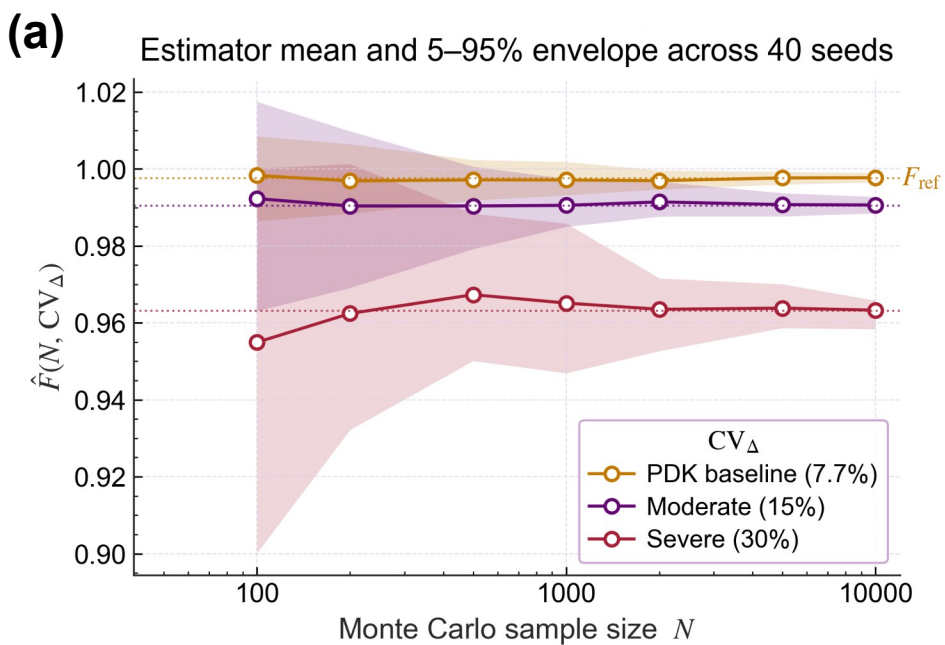


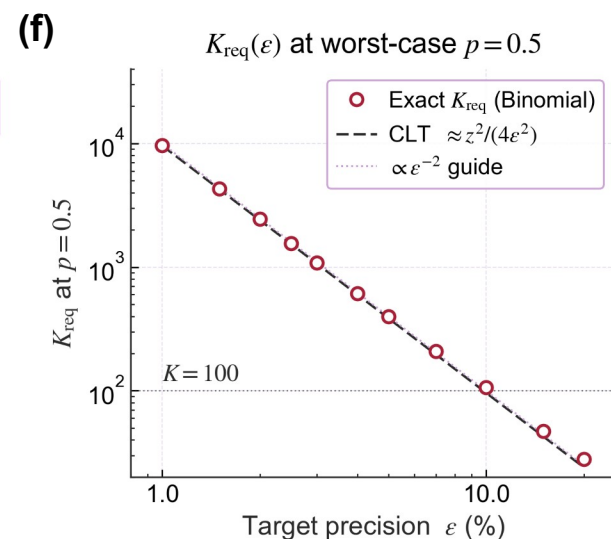
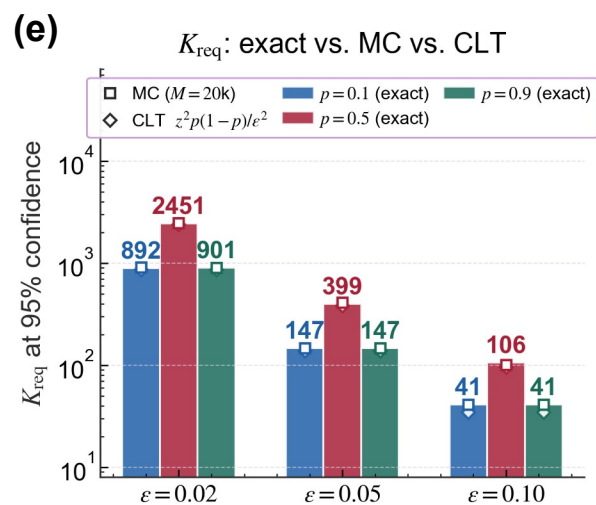
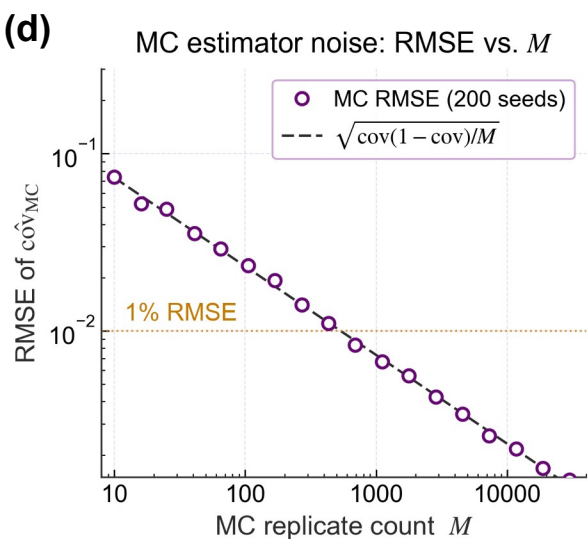
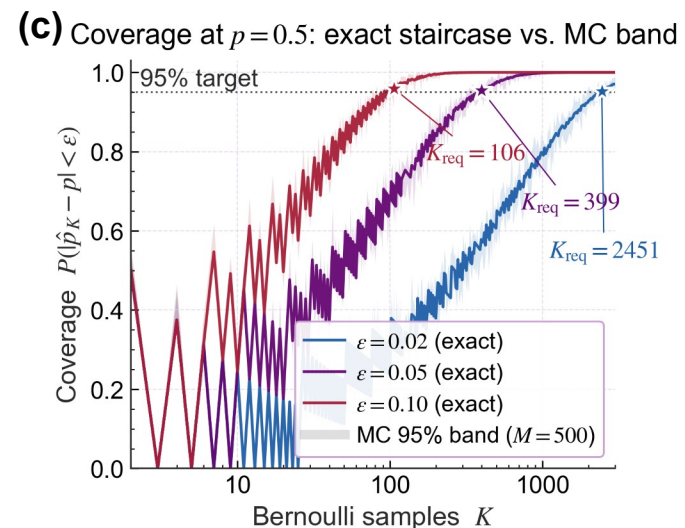
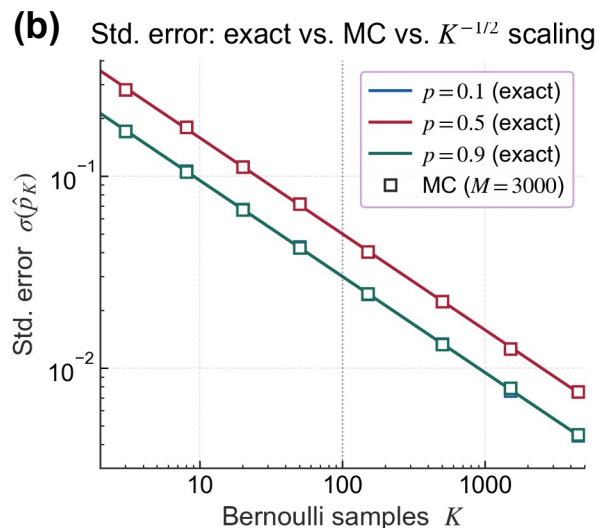
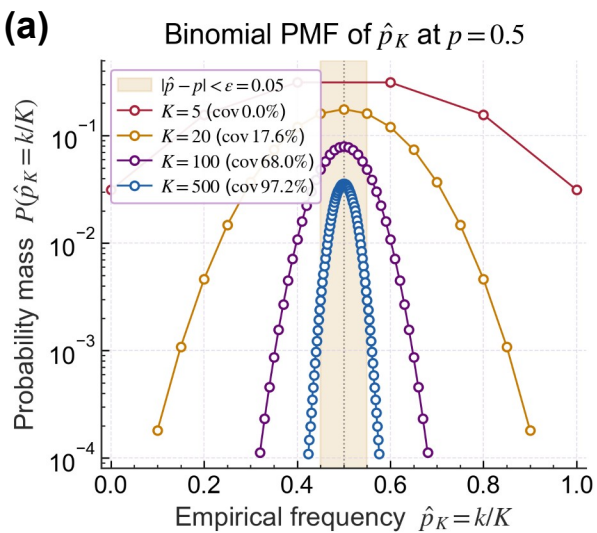


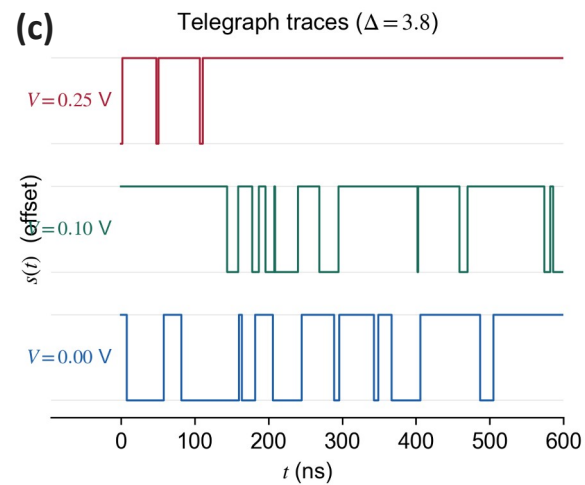
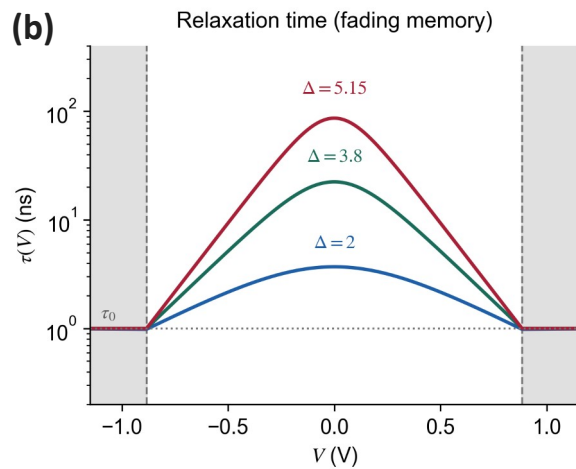
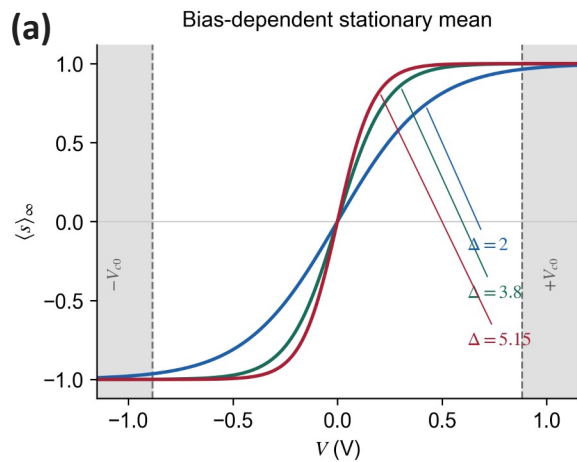


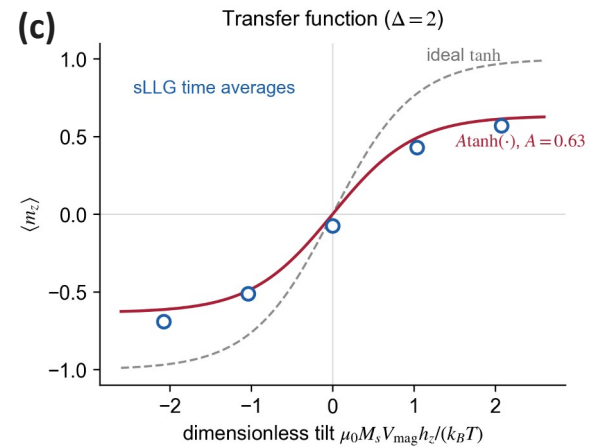
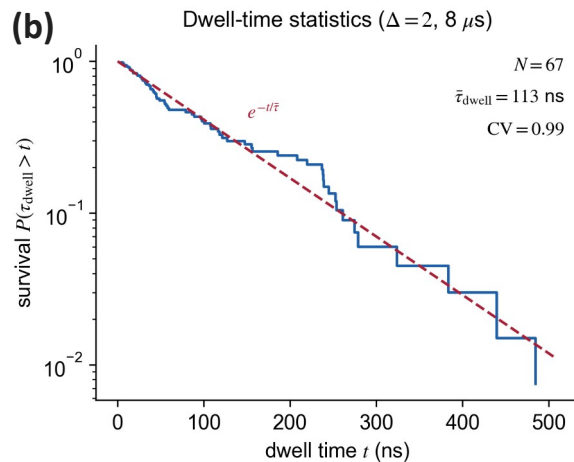
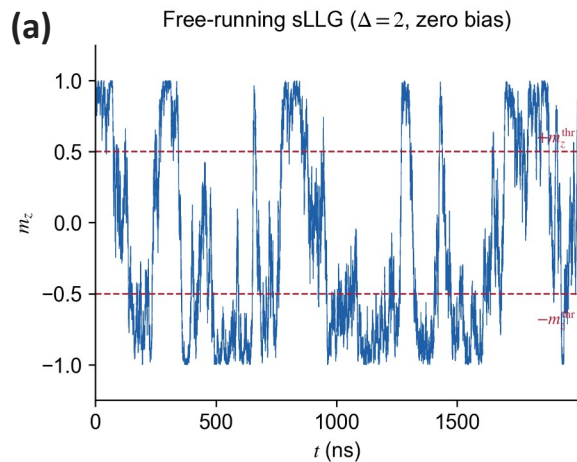
vg sot-sim process-variability Monte-Carlo P_{sw} sweep, $t_p = 0.75 \text{ ns}$, $V_{\text{MTJ}} = 0$



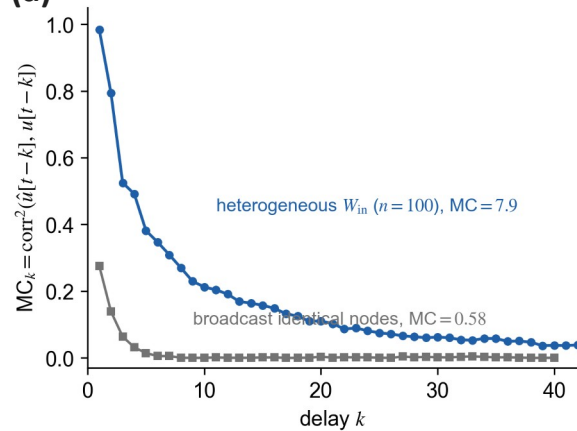




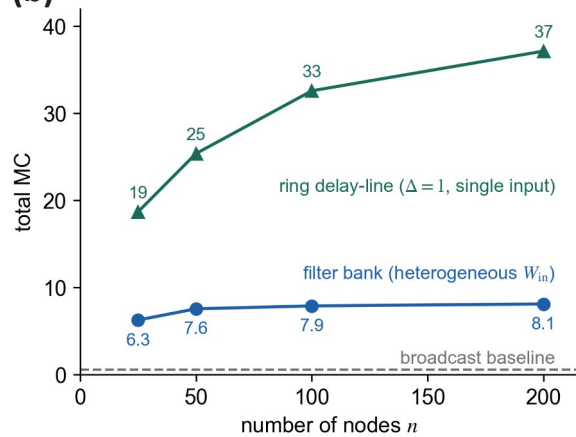




(a) Per-delay memory capacity



(b) Capacity vs node count



(c) Device-averaging cost ($n=100$)

